O-H-1934
June 2, 1975
SUPERSEDING
Fed. Spec. O-I-579
November 1, 1960 (see 6.4)

FEDERAL SPECIFICATION

HERBICIDE, SODIUM ARSENITE (SOLUTION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers an aqueous solution of sodium arsenite, containing 4 pounds of arsenic trioxide per gallon at 20°C (68°F)(see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-D-705 - Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

40 CFR 162-180 - Pesticide Programs.
49 CFR 170-179 - Hazardous Materials Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

D 1429 - Specific Gravity of Water and Waste Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Composition.

3.1.1 Arsenic trioxide. The solution shall contain a minimum of 32.0 percent by weight of arsenic trioxide, when tested as specified in 4.3.2.

3.1.2 Free alkali. The solution shall contain a minimum of 9.3 percent by weight of free alkali, as sodium hydroxide, when tested as specified in 4.3.3.

3.2 Specific gravity. The solution shall have a minimum specific gravity of 1.47 at 25°/25°C, when tested as specified in 4.3.4.

3.3 Registration and labeling. The herbicide shall be registered and labeled in compliance with the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), as amended. This act, and regulations promulgated thereunder, is administered by the Environmental Protection Agency (EPA).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspections. The sodium arsenite shall be inspected to determine conformance to the requirements of sections 3 and 5.

4.2.1 Quality conformance inspection sampling. Sampling for inspection shall be in accordance with MIL-STD-105, at the inspection levels and acceptable quality levels (AQL's) specified herein.

4.2.2 Sampling lot. A lot shall consist of not more than 10,000 U.S. gallons of solution or fraction thereof, prepared as one batch or unit, using the same raw materials and process.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one container, fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

Examine	Defects
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

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4.2.4 Testing of the end item. The sodium arsenite shall be tested to conform to the requirements of section 3, in accordance with the test methods specified in 4.3. The sample unit shall be a 1-quart composite, selected in equal portions from a lot (see 4.2.2) in accordance with inspection level S-2. The sample unit shall be thoroughly mixed to yield the composite sample, and stored in a clean, dry container with identifying lot number and date. The lot shall be unacceptable if the composite fails to meet any test requirement of this specification. The test report shall contain individual values utilized in expressing the final result.

4.3 Test methods.

4.3.1 Preparation of sample. Weigh 10 to 12g of the sodium arsenite solution to the nearest mg in a tared 500 ml volumetric flask. Dissolve in distilled water and make up to the volume. (Note that the sodium arsenite is highly toxic; handle with care.)

4.3.2 Arsenic trioxide. Remove a 25 ml aliquot from the prepared sample (see 4.3.1), transfer to a 250 ml Erlenmeyer flask, and add several drops of phenolphthalein solution. Neutralize by adding 0.1N hydrochloric acid dropwise until the pink coloration disappears. Add 40 ml of a cold 7 percent solution of sodium bicarbonate and titrate to a permanent blue color with standardized 0.1N iodine solution, using starch as the indicator. Calculate the percent arsenic trioxide as follows:

$$\text{Arsenic trioxide, percent} = \frac{V \times N \times 4.946}{W}$$

Where: V = milliliters of standard 0.1N iodine solution used.
N = normality of the iodine solution.
W = grams of sample (sodium arsenite) in the aliquot.

4.3.3 Free alkali. Remove a 25 ml aliquot from the prepared sample (see 4.3.1), transfer to a 125 ml Erlenmeyer flask, and add several drops of methyl purple indicator. Titrate with standardized 0.1N hydrochloric acid to the methyl purple end point. Calculate the percent free alkali (as NaOH) as follows:

$$\text{Free alkali, percent} = \frac{V \times N \times 4.0}{W}$$

Where: V = milliliters of standard 0.1N hydrochloric acid used.
N = normality of the hydrochloric acid solution.
W = grams of sample (sodium arsenite) in the aliquot.

4.3.4 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 1429, sections 6 to 9, except that the determination shall be made at 25 ± 1°C.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A and B. The insecticide shall be furnished in a 30-gallon drum conforming to PPP-D-705, type I.

5.1.2 Level C. The herbicide, in quantities as specified (see 6.2), shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Department of Transportation regulations relating to transportation (49 CFR 170-179).

5.2 Marking. In addition to any special marking required by the contract or order, containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified.

5.2.1 Labeling. Containers shall be labeled in compliance with 40 CFR 162-180, Pesticide Programs (see 3.3).

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6. NOTES

6.1 Intended use. Sodium arsenite is intended for use as a general herbicide to control all vegetation in fenced-in industrial areas or for prepaving treatment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity required.
(c) Level of packing required (see 5.1).
(d) Marking and labeling required (see 5.2 and 5.2.1).

6.3 Basis of purchase. Sodium arsenite solution shall be purchased by volume, the unit being a U.S. gallon at 25°C (77°F).

6.4 Supersession data. This specification includes the requirements of O-I-579, Insecticide, Sodium Arsenite Concentrate, dated November 1, 1960.

6.4.1 Change in concentration. The sodium arsenite concentration has been reduced from 6 pounds of arsenic trioxide per gallon to 4 pounds of arsenic trioxide per gallon.

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

EPA
GSA-FSS, PBO
JUS-FPI
NASA-LRC
USDA-ARS
VA-DMS

PREPARING ACTIVITY: GSA-FSS

Orders for this publication are to be placed with dministration, acting as an agent for the Superintendent of Document.. See section 2 of this specification to obtain extra copies and other documents referenced herein. cents each.

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